Cyclohexane Reactor Ver 1 · operator training simulator

Procurement packages

Document no. CHX-V1-PC-PKG-001 · **Discipline** Procurement

Client Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University (teaching)

• 17 procurement packages for the cyclohexane unit (benzene hydrogenation, 300 kt/y): reactors, exchangers, vessels, pumps, the recycle compressor, filters, valves, SIS, DCS, analysers, catalysts, piping, electrical, relief valves and fire and gas.
• Long-lead packages: PK-01 Slurry reactors R-101 (with agitator M-101) and R-102, PK-07 Recycle hydrogen compressor K-101. Critical path: PK-07; mechanical completion week 88 (indicative).
• Key data from the simulator at design, the Design tab and the Safety-layers assumptions; lead times and schedule durations are typical, not quotations.
• Companion document: pre-commissioning and commissioning plan CHX-V1-CM-PLN-001.

Revisions

Rev	Date	Description	Originator	Checker	Approver
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This document is generated for teaching from the simulator model of the unit. It is not issued for construction; every number comes from the model or is labelled assumed / typical practice. No client, contractor or vendor is implied.

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1 · Purpose and basis

This document defines the purchase packages of the unit: what each package contains, the data the supplier needs, the inspection and tests at the supplier's works (FAT) and at site (SAT), and an indicative procurement schedule. It is generated from the operator training simulator and is checked against it in the "Procurement to commissioning" tab.

• Packages cover the equipment, instruments and bulk of the screen P&ID (CHX-V1-PID-001) and the recommended additions of the Safety layers tab.
• Key data come from the simulator at design (noise off), the Design tab sizing and the Safety-layers assumptions; each number carries its source and is re-checked in the page.
• Lead times, PO weeks, shipping and installation durations are indicative and labelled typical: they teach the shape of a procurement schedule, not a quotation.
• Source codes of the key data: m: simulator measurement at design · p: model state at design · k: model constant (K) · ds: Design tab sizing · sl: Safety-layers assumption · pw: absorbed power from the model economics · alm: alarm table.
• Inspection marks: H hold point (witnessed by the purchaser, work stops until released), W witness (notified, may attend), R review of records, – not applicable.

Referenced documents

Document	Title / where
CHX-V1-PID-001	Screen P&ID (P&ID tab)
CHX-V1-PFD-001	Process flow diagram
CHX-V1-PE-HMB-001 / -LL-001	Heat and material balance; line list (Process engineering tab)
CHX-V1-ME-EQL-001, CHX-V1-ME-DS-<tag>	Equipment list and datasheets (Mechanical tab)
CHX-V1-PE-RLB-001	Relief load basis (Relief & vent tab)
CHX-V1-IN-INL-001, -CE-001, -SRS-001	Instrument list, C&E, SRS
CHX-V1-EE-LL-001, -HAC-001	Electrical load list, hazardous-area schedule
CHX-V1-CM-PLN-001	Pre-commissioning and commissioning plan

2 · Package register

Lead time = manufacture including FAT, ex works (typical range; the middle value is used in section 5). Float = mechanical-completion week − the week the package is installed.

Package	Title	Discipline	Tags	Lead, wk (typical)	PO wk	Delivered wk	Float, wk	Datasheet / document
PK-01 (long lead)	Slurry reactors R-101 (with agitator M-101) and R-102	Mechanical · static + rotating	R-101, M-101, R-102	52–60	8	70	8	CHX-V1-ME-DS-R-101, CHX-V1-ME-DS-R-102 (see the Mechanical tab)
PK-02	Trickle-bed finishing reactor R-103 with internals	Mechanical · static	R-103	36–44	14	60	22	CHX-V1-ME-DS-R-103 (see the Mechanical tab)
PK-03	Shell-and-tube exchangers E-101, E-103, E-106	Mechanical · static	E-101, E-103, E-106	30–40	14	56	26	CHX-V1-ME-DS-E-101, -E-103, -E-106 (see the Mechanical tab); E-106 duty from the HMB CHX-V1-PE-HMB-001 (Process engineering tab)
PK-04	Steam drum D-101	Mechanical · static	D-101	26–34	14	50	33	CHX-V1-ME-DS-D-101 (see the Mechanical tab)
PK-05	Pressure vessels V-103 and V-104	Mechanical · static	V-103, V-104	26–32	14	48	36	CHX-V1-ME-DS-V-103, -V-104 (see the Mechanical tab)
PK-06	Centrifugal pumps P-100A/B, P-101A/B, P-102A/B	Mechanical · rotating	P-100A, P-100B, P-101A, P-101B, P-102A, P-102B	30–40	16	58	26	CHX-V1-ME-DS-P-100, -P-101, -P-102 (see the Mechanical tab); hydraulics in the Process engineering tab
PK-07 (long lead)	Recycle hydrogen compressor K-101	Mechanical · rotating	K-101	60–72	8	80	0 critical	CHX-V1-ME-DS-K-101 (see the Mechanical tab)
PK-08	Catalyst candle filters F-102A/B	Mechanical · static	F-102A, F-102B	28–36	16	54	30	CHX-V1-ME-DS-F-102 (see the Mechanical tab)
PK-09	Control valves	Instrument	FV-101, FV-103, FV-104, PV-105, LV-101, LV-102, LV-104, LV-109, PV-110, TV-102, HV-107	20–30	20	52	33	Instrument datasheets with the instrument list CHX-V1-IN-INL-001 (see the Instrumentation & control tab)
PK-10	Safety instrumented system (logic solver, XV/EDV valves, SIS sensors)	Instrument · SIS	SIS-100, XV-101, XV-105, EDV-101, HS-100, TE-100, TE-107A…E, PT-101, LT-104S, AT-100	30–40	18	60	24	SRS CHX-V1-IN-SRS-001 (SRS tab), C&E CHX-V1-IN-CE-001 (C&E matrix tab)
PK-11	DCS and field instruments	Instrument · DCS	DCS, FIC-101, FIC-103, PIC-105, TIC-101, FIC-104, LIC-101, TIC-102, LIC-102, LIC-104, LIC-109, PIC-110, HIC-107	26–34	18	52	30	Instrument list CHX-V1-IN-INL-001 (see the Instrumentation & control tab)
PK-12	Process analysers AI-100, AI-103, AI-104, AI-105	Instrument · analysers	AI-100, AI-103, AI-104, AI-105	30–40	18	60	24	Analyser datasheets with CHX-V1-IN-INL-001 (see the Instrumentation & control tab)
PK-13	Catalysts: Raney-Ni (R-101/R-102) and supported Ni (R-103)	Process · catalyst	Raney-Ni, supported Ni	12–24	44	70	16	Catalyst specification with the process design basis (see the Process engineering tab)
PK-14	Piping bulk materials and manual valves	Piping	HV-110, HV-113, HV-116, HV-117, HV-120, HV-121	16–24	12	36	32	Line list CHX-V1-PE-LL-001 (Process engineering tab); piping material specification
PK-15	Relief and emergency isolation valves (recommended additions)	Instrument · relief	PSV-101, PSV-102, PSV-104, PSV-112, PSV-109, TSV-113, TSV-106, TSV-102J, TSV-102F, EIV-100, EIV-101, EIV-102, EIV-104, NRV-100, NRV-112	16–26	24	52	33	Relief load basis CHX-V1-PE-RLB-001 (see the Relief & vent tab); Safety layers tab

Package	Title	Discipline	Tags	Lead, wk (typical)	PO wk	Delivered wk	Float, wk	Datasheet / document
PK-16	Electrical: switchgear, MCC, motors interface, cables	Electrical	MCC, K-101 motor, M-101 motor, P-100A/B, P-101A/B, P-102A/B	26–40	16	56	24	Electrical load list CHX-V1-EE-LL-001 and hazardous-area schedule CHX-V1-EE-HAC-001 (see the Electrical tab)
PK-17	Fire and gas detection, fire water and safety equipment (recommended additions)	Safety · fire and gas	GD-1, GD-2, GD-3, FD-1, FW-1, HYD-1, HYD-2, MON-1, MON-2, DEL-101, DEL-104, MCP-1, SIREN, SHOWER	24–34	20	56	26	Safety layers tab (layer 6); fire and gas cause-and-effect to be added to the C&E

3 · Package scope, key data and tests

PK-01 · Slurry reactors R-101 (with agitator M-101) and R-102

Design, fabrication, inspection and test of the two slurry hydrogenation reactors: R-101 with its agitator M-101 (gas-dispersing impeller, gearbox, double mechanical seal with a barrier-fluid system — typical for hydrogen service), hydrogen sparger, catalyst-pot nozzle (HV-117), pump-around and overflow nozzles, level bridle and thermowells; R-102 with its heating/cooling jacket (MP steam or cooling water). Excludes instruments (PK-10, PK-11), insulation and the catalyst (PK-13).

Item	Value	Unit	Source
R-101 operating temperature	200	°C	m:TI100
R-101 operating pressure (PI-101)	30.0	bar g	m:PI101
R-101 liquid volume at 65 % level	29.0	m³	ds:Vl1
R-101 vessel volume (Design tab)	49.9	m³	ds:Vv1
R-101 diameter (H/D 1.2 assumed)	3.76	m	ds:D1
R-101 straight side	4.51	m	ds:Hs1
R-101 heat of reaction released	24.99	MW	ds:Qrx:0.001
Boiling point at reactor pressure (ceiling)	258	°C	ds:Tb
Agitator absorbed power at design (JI-101)	153	kW	m:JI101
R-102 operating temperature	160	°C	m:TI102
R-102 liquid volume at 55 % level	15.0	m³	ds:Vl2
R-102 vessel volume (Design tab)	30.0	m³	ds:Vv2
R-102 jacket heating duty at design	1.29	MW	ds:Q2j:0.001
Design pressure, hydrogen loop (assumed, = PSV set)	36	bar g	sl:Pdes
R-101 design temperature (assumed)	250	°C	sl:Tdes1

Inspection and test
Material certificates 3.1 and PMI of pressure parts; weld procedures and NDE to the code (RT/UT of main seams)
Shop hydrotest at 1.3 × design = 46.8 bar g (ASME VIII-1 UG-99, typical)
Agitator M-101 FAT: mechanical run of drive, gearbox and seal at rated speed (vibration, bearing temperatures, seal leakage); barrier-fluid system test
Dimensional check of nozzles and the sparger; final inspection and release note
Datasheet: CHX-V1-ME-DS-R-101, CHX-V1-ME-DS-R-102 (see the Mechanical tab)
Lead time 52–60 weeks (typical); PO week 8, shipping 6, installation 10 weeks (assumed)
Note: Long lead: thick-walled vessel for 36 bar g with an agitator and a seal system (typical 52–60 weeks).
Note: The model has an agitator motor on R-101 only (MOTORS); how R-102 keeps the catalyst suspended is EQ-ec-1.

PK-02 · Trickle-bed finishing reactor R-103 with internals

Vessel with liquid distributor tray, catalyst support grid, inert-ball layers, five thermowell cells for TE-107A…E (elements in PK-10), catalyst loading and dump nozzles. The supported-Ni catalyst is PK-13 and is loaded at site during pre-commissioning.

Item	Value	Unit	Source
Catalyst bed volume (LHSV 3 h^{-1} assumed)	16.2	m^3	ds:Vbed
Bed diameter (H/D 3 assumed)	1.90	m	ds:D3
Bed height	5.71	m	ds:H3
Liquid to the bed at 15 °C	48.7	m^3/h	ds:F3m3
Inlet temperature (TI-106)	158.5	°C	m:TI106
Bed temperature rise at design	0.08	K	ds:dT3
Rise per 1 % benzene slip	8.3	K	ds:dT3pc
Design pressure (assumed)	36	bar g	sl:Pdes
Design temperature (assumed)	260	°C	sl:Tdes3

Inspection and test
Material certificates and NDE
Shop hydrotest at 1.3 × design = 46.8 bar g (typical)
Distributor tray levelness and water distribution test (typical)
Internals trial fit; thermowell positions against the five cells
Datasheet: CHX-V1-ME-DS-R-103 (see the Mechanical tab)
Lead time 36–44 weeks (typical); PO week 14, shipping 6, installation 6 weeks (assumed)

PK-03 · Shell-and-tube exchangers E-101, E-103, E-106

E-101 pump-around cooler and steam generator (pump-around on the tubes, boiling water from D-101 on the shell; MP steam on the shell for start-up heating), E-103 reflux condenser on cooling water, E-106 product cooler on cooling water. TEMA class R construction (typical for this service).

Item	Value	Unit	Source
E-101 duty at design	9.60	MW	p:Qe1:0.001
E-101 area (U 0.85 kW/m^2K assumed)	706	m^2	ds:A1
E-101 pump-around flow (FIC-104)	357	t/h	m:FI104
E-101 pump-around return (TI-104)	159.3	°C	m:TI104
E-101 shell: D-101 pressure (PIC-110)	4.5	bar g	m:PI110
E-101 shell boiling temperature	155.8	°C	ds:Tsat
E-103 duty	7.85	MW	ds:Qc:0.001
E-103 cooling-water outlet	47.6	°C	ds:Tcwo
E-103 vent temperature (TI-103)	55.7	°C	m:TI103
E-106 inlet (TI-106)	158.5	°C	m:TI106
E-106 outlet = V-104 (TI-109)	44.9	°C	m:TI109
Cooling-water supply	32	°C	k:Tcw0
Tube side design pressure (assumed)	36	bar g	sl:Pdes
Cooling-water side relief set (assumed)	10	bar g	sl:Tsv

Inspection and test
Material certificates, PMI; tube-to-tubesheet weld or expansion procedure
Hydrotest of each side separately; tube-side test with the shell open for inspection
Tube-to-tubesheet joint leak test of E-101 with helium (hydrocarbon/boiler-water boundary; typical)
Bundle pull test and dimensional check
Datasheet: CHX-V1-ME-DS-E-101, -E-103, -E-106 (see the Mechanical tab); E-106 duty from the HMB CHX-V1-PE-HMB-001 (Process engineering tab)
Lead time 30–40 weeks (typical); PO week 14, shipping 6, installation 6 weeks (assumed)
Note: The E-101 shell sees MP steam at 10 bar g in HEAT mode while the D-101 relief is assumed at 8 bar g: shell design pressure is EQ-ec-2.

PK-04 · Steam drum D-101

Horizontal steam drum on the E-101 shell side (natural or pump-assisted circulation per the Mechanical tab) with feed-water distributor, steam separators and demister, continuous and intermittent blowdown nozzles, level bridle for LIC-109 and gauge glass.

Item	Value	Unit	Source
Operating pressure (PIC-110)	4.5	bar g	m:PI110
LP steam export (FI-111)	14.7	t/h	m:FI111
BFW flow (FI-109)	14.9	t/h	m:FI109
BFW temperature	105	°C	k:Tbfw
Normal level (LIC-109)	50	%	m:LI109
Safety valve PSV-109 set (assumed, recommended SL-R12)	8.0	bar g	sl:Psv109

Inspection and test
Material certificates and NDE
Shop hydrotest at 1.3 × design (typical)
Internals inspection (separators, distributor)
Datasheet: CHX-V1-ME-DS-D-101 (see the Mechanical tab)
Lead time 26–34 weeks (typical); PO week 14, shipping 6, installation 5 weeks (assumed)

PK-05 · Pressure vessels V-103 and V-104

V-103 reflux drum below E-103 (condensed cyclohexane back to R-101); V-104 high-pressure separator after E-106 with a mesh demister (protects K-101), product nozzle to LV-104, separate nozzles for LT-104 and the SIS level element LT-104S, make-up hydrogen inlet. The Raney-Ni catalyst pot on R-101 (HV-117) is in this package; it has no equipment number on the P&ID (EQ-ec-3).

Item	Value	Unit	Source
V-103 reflux flow (FI-105)	46.5	t/h	m:FI105
V-103 temperature (TI-103)	55.7	°C	m:TI103
V-104 pressure (PIC-105)	29.0	bar g	m:PI105
V-104 temperature (TI-109)	44.9	°C	m:TI109
V-104 normal level (LIC-104)	50	%	m:LI104
Product flow (FI-110)	37.5	t/h	m:FI110
LSHH-104 setting (K-101 stop)	90	%	k:L4hh
Design pressure (assumed)	36	bar g	sl:Pdes

Inspection and test
Material certificates and NDE
Shop hydrotest at 1.3 × design = 46.8 bar g (typical)
Demister installation check
Datasheet: CHX-V1-ME-DS-V-103, -V-104 (see the Mechanical tab)
Lead time 26–32 weeks (typical); PO week 14, shipping 6, installation 4 weeks (assumed)

PK-06 · Centrifugal pumps P-100A/B, P-101A/B, P-102A/B

Six API 610 process pumps (typical) with motors, seals and seal-support plans, couplings and baseplates: P-100A/B benzene feed to R-101, P-101A/B hot slurry pump-around through E-101 (hardened wetted parts, flushed seal — typical for catalyst slurry), P-102A/B R-102 slurry through F-102 to R-103. Includes the minimum-flow bypasses of HAZOP R35 and the discharge check valves of R08 / SL-R21 where the P&ID adds them.

Item	Value	Unit	Source
P-100 benzene flow (FI-101)	34.8	t/h	m:FI101
P-100 absorbed power (model)	90	kW	pw:P100
P-101 pump-around flow (FIC-104)	357	t/h	m:FI104
P-101 pumping temperature (TI-100)	200	°C	m:TI100
P-101 differential pressure (model)	3.0	bar	k:pumpDP
P-101 absorbed power (model)	160	kW	pw:P101
P-102 flow R-102 → R-103 (FI-102)	20.9	t/h	m:FI102
P-102 pumping temperature (TI-102)	160	°C	m:TI102
P-102 absorbed power (model)	30	kW	pw:P102

Inspection and test
Casing hydrotest at 1.5 × MAWP (API 610, typical)
Performance test on water: Q–H, power, efficiency at five points; NPSHr test for P-101 (hot slurry service)
Mechanical running test 4 h (vibration, bearing temperatures, seal leakage; typical)
Motor routine test certificates (PK-16 specification)
Datasheet: CHX-V1-ME-DS-P-100, -P-101, -P-102 (see the Mechanical tab); hydraulics in the Process engineering tab
Lead time 30–40 weeks (typical); PO week 16, shipping 6, installation 4 weeks (assumed)

PK-07 · Recycle hydrogen compressor K-101

Recycle compressor package for the hydrogen loop: compressor (type selected at bid; centrifugal per API 617 assumed for a low pressure ratio), gearbox if needed, driver motor, lube-oil and seal (dry-gas seal assumed) systems, anti-surge / kickback valve and controls (HAZOP R27), discharge check valve (SL-R22), local panel. Suction from V-104; liquid protection by LSHH-104 (C&E).

Item	Value	Unit	Source
Flow (FIC-103, recycle + make-up)	46,000	Nm³/h	m:FI103
Suction pressure (V-104, PIC-105)	29.0	bar g	m:PI105
Discharge pressure (PI-112)	32.0	bar g	m:PI112
Pressure ratio (absolute)	1.100	–	authored / assumed text
Gas hydrogen purity (AI-105)	92.1	%	m:AI105
Absorbed power (model)	1,200	kW	pw:K101
Discharge piping design pressure (assumed)	42	bar g	sl:PdesK
Start interlock: suction pressure at least	10	bar g	authored / assumed text
Start interlock: V-104 level at most	85	%	authored / assumed text

Inspection and test
Casing hydrotest; impeller overspeed test and rotor balance (API 617, typical)
Mechanical running test 4 h at maximum continuous speed: vibration, bearing temperatures, seal-gas leakage (typical)
Lube-oil and seal-gas system FAT; helium leak test of the casing for hydrogen service (typical)
Optional performance test (ASME PTC 10 type 2 on a substitute gas)
Datasheet: CHX-V1-ME-DS-K-101 (see the Mechanical tab)
Lead time 60–72 weeks (typical); PO week 8, shipping 6, installation 8 weeks (assumed)
Note: Long lead and the critical path of this plan (part 4). Low-molecular-weight gas: the machine is sized on hydrogen; a run on nitrogen at site needs reduced speed or suction pressure (vendor limit) — see the motor solo-run notes.

PK-08 · Catalyst candle filters F-102A/B

Two pressure candle filters (one running, one on stand-by) for the R-102 slurry ahead of R-103, with back-flush connections, wet cake discharge into closed drums (SL-R27), change-over valves and a thermal relief on each (SL-R16).

Item	Value	Unit	Source
Flow through the running filter (FI-102)	20.9	t/h	m:FI102
Temperature (TI-102)	160	°C	m:TI102
Clean ΔP at design (PDI-102)	0.26	bar	m:PDI102
ΔP alarm (PDAH-102)	0.8	bar	alm:PDI-102:hi
Design pressure (assumed)	36	bar g	sl:Pdes

Inspection and test
Shop hydrotest (typical 1.3 × design)
Candle integrity (bubble-point) test of each element (typical)
Change-over and back-flush valve function test
Datasheet: CHX-V1-ME-DS-F-102 (see the Mechanical tab)
Lead time 28–36 weeks (typical); PO week 16, shipping 6, installation 4 weeks (assumed)

PK-09 · Control valves

One control valve per loop of the DCS (table below) with actuator, positioner, solenoid where the fail action needs it, and limit switches. FV-104 is a three-way valve in hot catalyst slurry (wear-resistant trim, typical). Hand valves HV-110/113/116/117/120/121 are piping bulk (PK-14).

Item	Value	Unit	Source
Benzene flow through FV-101	34.8	t/h	m:FI101
Hydrogen through FV-103	46,000	Nm³/h	m:FI103
Pump-around through FV-104	357	t/h	m:FI104
Make-up through PV-105 (FI-106)	31,974	Nm³/h	m:FI106
Purge through HV-107 (FI-107)	2,017	Nm³/h	m:FI107
LP steam through PV-110 (FI-111)	14.7	t/h	m:FI111

Inspection and test
Shell hydrotest (typical 1.5 × rating)
Seat leakage test: class IV typical; class V for PV-105 and LV-104 (gas blow-by service, assumed)
Stroke test, fail position on air loss and on signal loss, positioner calibration 0-25-50-75-100 %
Cv / characteristic certificate
Datasheet: Instrument datasheets with the instrument list CHX-V1-IN-INL-001 (see the Instrumentation & control tab)
Lead time 20–30 weeks (typical); PO week 20, shipping 6, installation 3 weeks (assumed)

Loop	Valve	Service	Fail action	Set point
FIC-101	FV-101	Benzene feed to R-101 → FV-101	fail closed	34.8 t/h
FIC-103	FV-103	Hydrogen (recycle + make-up) to R-101 sparger → FV-103	fail closed	46,000 Nm³/h
PIC-105	PV-105	Loop pressure at V-104 → make-up hydrogen PV-105	fail closed	29.0 bar g
TIC-101	FV-104 (cascade)	R-101 temperature → cascade to FIC-104 pump-around through E-101	via FIC-104	200.0 °C
FIC-104	FV-104	Pump-around flow through E-101 → FV-104	fail open	357.3 t/h
LIC-101	LV-101	R-101 level → LV-101 overflow to R-102	fail closed	65.0 %
TIC-102	TV-102	R-102 temperature → TV-102 jacket (MP steam in HEAT, CW in COOL)	fail closed	160.0 °C
LIC-102	LV-102	R-102 level → LV-102 to R-103 through filter F-102	fail closed	55.0 %
LIC-104	LV-104	V-104 separator level → LV-104 product to storage	fail closed	50.0 %
LIC-109	LV-109	D-101 steam drum level → LV-109 boiler feed water	fail open	50.0 %
PIC-110	PV-110	D-101 steam pressure → PV-110 LP steam export	fail open	4.5 bar g
HIC-107	HV-107	Purge from V-104 to fuel gas (hand controller)	fail closed	25.0 %

PK-10 · Safety instrumented system (logic solver, XV/EDV valves, SIS sensors)

Logic solver SIS-100 (separate from the DCS, SIL 3 capable assumed) with first-out and latching; final elements XV-101 (benzene), XV-105 (make-up hydrogen) — tight shut-off, fail closed through a solenoid, fire-safe — and EDV-101 (depressuring to flare, fail open); SIS sensors with their own transmitters: TE/TT-100, TE-107A…E, PT-101, LT-104S, AT-100 (oxygen permissive); ESD pushbutton HS-100; run-status contacts of M-101 and K-101 (XS-101, XS-102) from the MCC. Allow spare I/O and valves for the open recommendations that add SIS scope (EQ-ec-4).

Item	Value	Unit	Source
C&E trip rows (incl. ESD)	7	rows	authored / assumed text
C&E permissive rows	10	rows	authored / assumed text
TSHH-100 setting	220	°C	k:T1hh
TSHH-107 setting	220	°C	k:T3hh
PSHH-101 setting	33.5	bar g	k:P1hh
LSHH-104 setting	90	%	k:L4hh
Oxygen permissive AI-100 below	1	%	k:O2max

Inspection and test
SIS FAT against the C&E: every cause row forced at the logic-solver input, every marked effect, delay, latch, first-out and reset rule checked (part 7 lists the rows)
XV-101 / XV-105 seat leakage class VI (tight shut-off, typical for SIS isolation of gas) and stroke time ≤ 2 s (SRS proof-test criterion); EDV-101 stroke to open
Fire-safe type test certificate (API 607, typical); SIL capability certificates (IEC 61508) for sensors, logic solver and valves
Integrated FAT with the DCS (status read-back, first-out display)
Datasheet: SRS CHX-V1-IN-SRS-001 (SRS tab), C&E CHX-V1-IN-CE-001 (C&E matrix tab)
Lead time 30–40 weeks (typical); PO week 18, shipping 6, installation 4 weeks (assumed)

PK-11 · DCS and field instruments

Distributed control system (controllers, I/O, operator and engineering stations, historian) and the field transmitters, thermowells and indicators of the BPCS: 12 control loops and the alarm table of the page. Graphics follow the screen P&ID.

Item	Value	Unit	Source
Control loops (makeCtrls)	12	loops	authored / assumed text
Alarm limits armed (C&E alarm rows)	28	limits	authored / assumed text
TIC-101 set point	200	°C	m:TIC101
PIC-105 set point	29.0	bar g	m:PI105

Inspection and test
DCS FAT: 100 % I/O check, every loop configured as the instrument list (range, action, tuning), alarm limits against the alarm table, graphics against the P&ID
Integrated FAT with SIS-100
Transmitter calibration certificates (5-point)
Datasheet: Instrument list CHX-V1-IN-INL-001 (see the Instrumentation & control tab)
Lead time 26–34 weeks (typical); PO week 18, shipping 4, installation 6 weeks (assumed)

PK-12 · Process analysers AI-100, AI-103, AI-104, AI-105

Analyser shelter with sample systems: AI-100 loop oxygen (also the hydrogen-admission permissive, AT-100 in PK-10), AI-103 product benzene and AI-104 MCP isomer (process gas chromatograph, typical), AI-105 recycle-hydrogen purity. Calibration gases and a lab sample point at each.

Item	Value	Unit	Source
AI-103 benzene at design	2.0	ppm	m:AI103
AI-103 specification (alarm)	10	ppm	alm:AI-103:hi
AI-104 MCP at design	60	ppm	m:AI104
AI-104 specification (alarm)	200	ppm	alm:AI-104:hi
AI-105 purity at design	92.1	%	m:AI105
AI-105 low alarm	85	%	alm:AI-105:lo
AI-100 permissive limit	1	%	k:O2max

Inspection and test
FAT with zero and span gases; repeatability and response time; shelter HVAC and gas detection
SAT with certified calibration gases; comparison with laboratory results
Datasheet: Analyser datasheets with CHX-V1-IN-INL-001 (see the Instrumentation & control tab)
Lead time 30–40 weeks (typical); PO week 18, shipping 6, installation 4 weeks (assumed)

PK-13 · Catalysts: Raney-Ni (R-101/R-102) and supported Ni (R-103)

Raney-nickel powder, first charge, supplied wet under liquid in sealed drums (pyrophoric when dry; carrier liquid to be agreed, EQ-ec-5); supported-nickel catalyst for R-103 (pre-reduced and passivated, typical) with inert ceramic balls for the bed. Quantities on the activity basis of the process design; the model represents activity only (cat = 1.0 at design).

Item	Value	Unit	Source
R-103 bed volume	16.2	m³	ds:Vbed
R-101 catalyst activity in the model at design	1.0	–	k:cat0
R-101 conversion at design (Design tab)	98	%	ds:X1:100

Inspection and test
Certificate of analysis per lot (Ni content, particle size, activity test — typical)
Receipt inspection: every Raney-Ni drum shows liquid above the catalyst; drums not bulging
Supported Ni: crush strength and loss on attrition sample (typical)
Datasheet: Catalyst specification with the process design basis (see the Process engineering tab)
Lead time 12–24 weeks (typical); PO week 44, shipping 6, installation 2 weeks (assumed)
Note: Ordered late on purpose: the catalyst should arrive shortly before loading, not age in storage (typical practice).

PK-14 · Piping bulk materials and manual valves

Pipe, fittings, flanges, gaskets, bolting, manual valves (hand valves HV-110, HV-113, HV-116, HV-117, HV-120, HV-121), strainers, spectacle blinds and the temporary items for flushing and testing (spools, blinds, target plates). Hydrogen-loop materials checked against API 941 (typical); flange class 300 for the loop assumed — confirm against the line list CHX-V1-PE-LL-001.

Item	Value	Unit	Source
Hydrogen loop design pressure (assumed)	36	bar g	sl:Pdes
K-101 discharge design pressure (assumed)	42	bar g	sl:PdesK
R-101 temperature (highest process line)	200	°C	m:TI100
LP steam pressure (D-101)	4.5	bar g	m:PI110

Inspection and test
Material certificates and PMI of alloy items
Valve shell and seat tests (API 598, typical)
Site: test packs, NDE and the hydrotest records of part 6
Datasheet: Line list CHX-V1-PE-LL-001 (Process engineering tab); piping material specification
Lead time 16–24 weeks (typical); PO week 12, shipping 4, installation 20 weeks (assumed)

PK-15 · Relief and emergency isolation valves (recommended additions)

Safety and thermal relief valves, emergency isolation valves and check valves added by the Safety-layers tab (SL-R8 … SL-R22). Orifice sizes from the relief load basis CHX-V1-PE-RLB-001 (Relief & vent tab); this package cannot be bought before that basis is issued (EQ-ec-6).

Item	Value	Unit	Source
PSV-101/102/104 set (assumed)	36	bar g	sl:Pdes
PSV-112 set (assumed)	42	bar g	sl:PdesK
PSV-109 set (assumed)	8.0	bar g	sl:Psv109
Thermal relief set, CW sides (assumed)	10	bar g	sl:Tsv

Inspection and test
Set-pressure test and seat tightness (API 527, typical) with certificate and seal
EIV: fire-safe certificate (API 607), stroke and fail position
Check valves: back-flow seat test
Datasheet: Relief load basis CHX-V1-PE-RLB-001 (see the Relief & vent tab); Safety layers tab
Lead time 16–26 weeks (typical); PO week 24, shipping 6, installation 3 weeks (assumed)

PK-16 · Electrical: switchgear, MCC, motors interface, cables

Medium-voltage starter for the K-101 motor (typical above about 300 kW), LV motor control centre for the pumps and the agitator, run-status contacts and stop relays for the SIS, power and control cables, local control stations, earthing, lighting, and hazardous-area equipment per CHX-V1-EE-HAC-001. Motors are supplied with the driven equipment to the electrical specification.

Item	Value	Unit	Source
Total absorbed power at design (model, econ)	1,633	kW	pw:total
K-101	1,200	kW	pw:K101
M-101	153	kW	pw:M101
P-100 (one running)	90	kW	pw:P100
P-101 (one running)	160	kW	pw:P101
P-102 (one running)	30	kW	pw:P102

Inspection and test
Switchgear and MCC routine tests (IEC 61439, typical)
Motor routine test certificates (IEC 60034, typical)
Site: insulation resistance and continuity of every cable; protection relay settings; motor solo runs (part 6)
Datasheet: Electrical load list CHX-V1-EE-LL-001 and hazardous-area schedule CHX-V1-EE-HAC-001 (see the Electrical tab)
Lead time 26–40 weeks (typical); PO week 16, shipping 6, installation 8 weeks (assumed)

PK-17 · Fire and gas detection, fire water and safety equipment (recommended additions)

Fire-and-gas panel with hydrogen detectors at K-101 and the R-101 top flanges, a hydrocarbon detector at the benzene pumps and a UV/IR flame detector over K-101 and V-104; fire-water ring main, hydrants, monitors, deluge on K-101 and water spray on V-104; manual call points, siren, windsock, muster sign and the safety shower. All are recommended additions of the Safety layers tab (SL-R29 ... SL-R44) and HAZOP R25.

Item	Value	Unit	Source
Gas detectors (SL-R36 ... SL-R38)	3	detectors	authored / assumed text
Flame detectors (SL-R39)	1	detector	authored / assumed text
Gas alarm levels (assumed)	10 / 25	% LFL	authored / assumed text

Inspection and test
Fire-and-gas panel FAT with the cause-and-effect of the detectors (typical)
Detector calibration with certified test gas at site; flame-detector test lamp
Fire-water ring main hydrotest and flushing; deluge valve trip test (typical)
Datasheet: Safety layers tab (layer 6); fire and gas cause-and-effect to be added to the C&E
Lead time 24–34 weeks (typical); PO week 20, shipping 6, installation 6 weeks (assumed)

4 · Inspection and test plan (FAT / SAT matrix)

H hold point: witnessed by the purchaser, work stops until released · **W** witness point · **R** review of records · – not applicable.

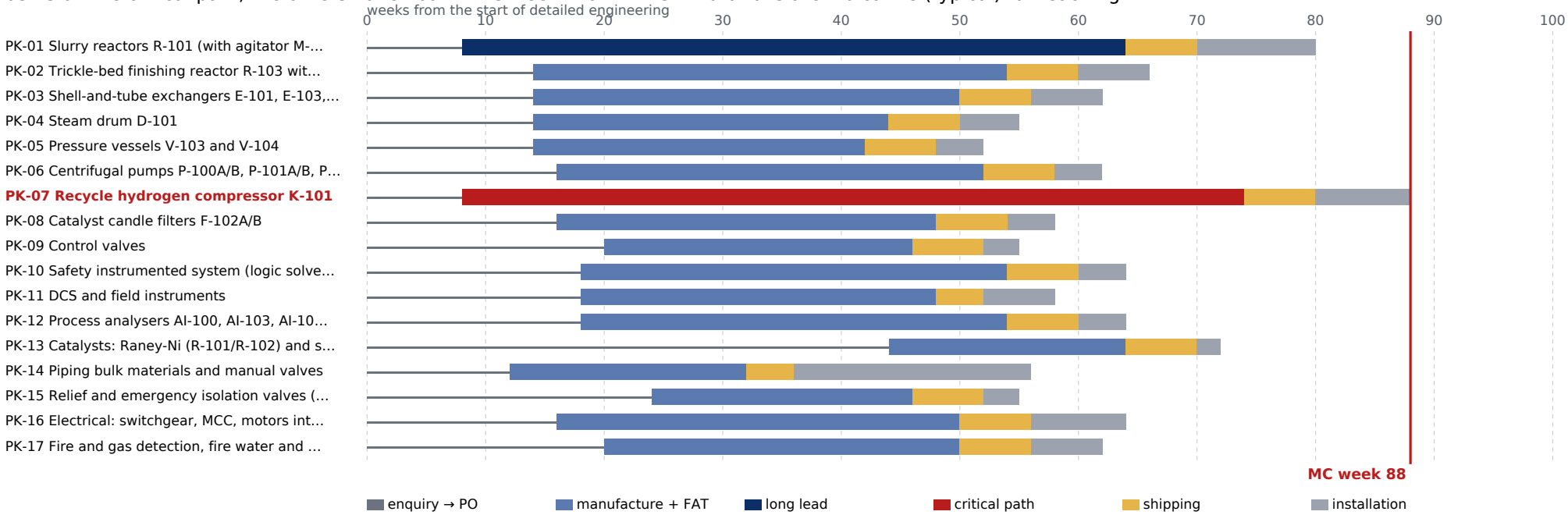
Package	MDR	HT	MRT	PERF	VLV	SISF	DCSF	SAT
PK-01 Slurry reactors R-101 (with agitator M-101) and R-102	H	H	H	W	–	–	–	W
PK-02 Trickle-bed finishing reactor R-103 with internals	H	H	–	W	–	–	–	R
PK-03 Shell-and-tube exchangers E-101, E-103, E-106	H	H	–	–	–	–	–	R
PK-04 Steam drum D-101	H	H	–	–	–	–	–	R
PK-05 Pressure vessels V-103 and V-104	H	H	–	–	–	–	–	R
PK-06 Centrifugal pumps P-100A/B, P-101A/B, P-102A/B	R	H	H	H	–	–	–	W
PK-07 Recycle hydrogen compressor K-101	H	H	H	W	–	–	–	W
PK-08 Catalyst candle filters F-102A/B	R	H	–	W	W	–	–	W
PK-09 Control valves	R	W	–	–	H	–	–	W
PK-10 Safety instrumented system (logic solver, XV/EDV valves, SIS sensors)	R	W	–	–	H	H	W	H
PK-11 DCS and field instruments	R	–	–	–	–	W	H	H
PK-12 Process analysers AI-100, AI-103, AI-104, AI-105	R	–	–	H	–	–	W	H
PK-13 Catalysts: Raney-Ni (R-101/R-102) and supported Ni (R-103)	H	–	–	R	–	–	–	–
PK-14 Piping bulk materials and manual valves	R	–	–	–	W	–	–	–
PK-15 Relief and emergency isolation valves (recommended additions)	R	H	–	–	H	–	–	W
PK-16 Electrical: switchgear, MCC, motors interface, cables	R	–	–	–	–	–	–	H
PK-17 Fire and gas detection, fire water and safety equipment (recommended additions)	R	W	–	W	W	–	W	H

Code	Test	Basis
MDR	Material certificates, PMI, WPS/PQR, NDE records	EN 10204 3.1; code of construction (typical)
HT	Shop hydrostatic test	ASME VIII-1 UG-99 1.3 × MAWP; API 610 / 617 casings 1.5 × (typical)
MRT	Mechanical running test	API 610 / 617 style: vibration, bearing temperatures, seal leakage (typical)
PERF	Performance test	Pump Q–H–power–NPSHr curve; compressor or agitator duty point (typical)
VLV	Valve tests: shell, seat leakage class, stroke and fail position	IEC 60534-4 / FCI 70-2 leak classes; API 598; API 527 for PSVs (typical)
SISF	SIS FAT against the C&E (every cause row)	IEC 61511-1 §13 (FAT), C&E CHX-V1-IN-CE-001
DCSF	DCS FAT: I/O, loops, graphics, alarm limits	Against the instrument list and the alarm table (typical)
SAT	Site acceptance / function test after installation	Same scope repeated with the field devices (typical)

The tests this unit cannot do without: the K-101 mechanical running test (hydrogen recycle, the critical-path machine), the pump performance tests (P-101 hot slurry service), the SIS FAT against every cause row of the C&E CHX-V1-IN-CE-001, and the seat-leakage class of the valves that isolate hydrogen and benzene (XV-101, XV-105, PV-105, LV-104).

5 · Procurement schedule and critical path

Each bar is PO week + manufacture and FAT + shipping + installation. Mechanical completion (MC) cannot come before the latest bar ends: the package that ends last is on the critical path; the others have float = MC week − own finish. Durations are indicative (typical) for teaching.



Worked example. K-101 (PK-07): PO week 8 + lead 66 + shipping 6 = delivered week 80; + installation 8 = week 88 = MC (critical, float 0). R-101/R-102 (PK-01): 8 + 56 + 6 = 70; + 10 = 80; float = 88 − 80 = **8 weeks**. A six-week slip of K-101 moves MC by six weeks; the same slip of PK-01 is absorbed by its float.

Package	PO wk	Lead	Ship	Delivered	Install	Installed	Float	Critical
PK-01 Slurry reactors R-101 (with agitator M-101) and R-102	8	56	6	70	10	80	8	
PK-02 Trickle-bed finishing reactor R-103 with internals	14	40	6	60	6	66	22	
PK-03 Shell-and-tube exchangers E-101, E-103, E-106	14	36	6	56	6	62	26	
PK-04 Steam drum D-101	14	30	6	50	5	55	33	
PK-05 Pressure vessels V-103 and V-104	14	28	6	48	4	52	36	
PK-06 Centrifugal pumps P-100A/B, P-101A/B, P-102A/B	16	36	6	58	4	62	26	
PK-07 Recycle hydrogen compressor K-101	8	66	6	80	8	88	0	yes
PK-08 Catalyst candle filters F-102A/B	16	32	6	54	4	58	30	
PK-09 Control valves	20	26	6	52	3	55	33	
PK-10 Safety instrumented system (logic solver, XV/EDV valves, SIS sensors)	18	36	6	60	4	64	24	

Package	PO wk	Lead	Ship	Delivered	Install	Installed	Float	Critical
PK-11 DCS and field instruments	18	30	4	52	6	58	30	
PK-12 Process analysers AI-100, AI-103, AI-104, AI-105	18	36	6	60	4	64	24	
PK-13 Catalysts: Raney-Ni (R-101/R-102) and supported Ni (R-103)	44	20	6	70	2	72	16	
PK-14 Piping bulk materials and manual valves	12	20	4	36	20	56	32	
PK-15 Relief and emergency isolation valves (recommended additions)	24	22	6	52	3	55	33	
PK-16 Electrical: switchgear, MCC, motors interface, cables	16	34	6	56	8	64	24	
PK-17 Fire and gas detection, fire water and safety equipment (recommended additions)	20	30	6	56	6	62	26	

6 · Scope added by the HAZOP and Safety-layers recommendations

73 recommendations add scope to a package. They must be closed (or the scope confirmed) before the package is ordered; the SIS package depends on the open SIS recommendations (EQ-ec-4).

ID	Source	Recommendation	Disposition	Package
R01	HAZOP	Add a high-high benzene flow or benzene-to-hydrogen ratio trip (FI-101 against FI-103) that closes XV-101; the feed can run far above design with no trip.	SIS	PK-10
R02	HAZOP	Confirm and document the pressure relief of R-101, R-102, R-103 and V-104 for blocked outlet, make-up valve wide open and fire; relief valves are not shown on the training P&ID.	DOC	PK-15
R03	HAZOP	Add a high-high temperature trip on R-102 with its own element (set below the R-102 boiling point at loop pressure).	SIS	PK-10
R04	HAZOP	Add low-low agitator power (JI-101) to the SIS so that a running motor with a failed drive is seen like a stopped motor.	SIS	PK-10
R05	HAZOP	Add an independent R-101 level element with high-high and low-low actions (stop feed; protect P-101A/B).	SIS	PK-10
R06	HAZOP	Add low cooling-water flow on E-103 or high-high TI-103 to stop the benzene feed.	SIS	PK-10
R07	HAZOP	Add independent high-high and low-low level on R-102 (stop P-102A/B, stop feed).	SIS	PK-10
R08	HAZOP	Fit a non-return valve on the P-100A/B discharge and close XV-101 when no feed pump runs.	HW	PK-06
R11	HAZOP	Add a low-flow alarm on the pump-around FI-104 (none in the alarm list).	SIS	PK-11
R13	HAZOP	Confirm the D-101 steam-side relief for blocked export at maximum E-101 duty.	DOC	PK-15
R14	HAZOP	Add low-low and high-high level on D-101 independent of LIC-109.	SIS	PK-10
R16	HAZOP	Interlock HV-110 (MP steam to E-101) closed whenever XV-101 is open, and put the HEAT to COOL switch-over in the light-off steps.	SIS	PK-10
R17	HAZOP	Detect hydrogen and hydrocarbon in the CW return of E-103, E-106 and the R-102 jacket (degassing pot or analyser).	HW	PK-12
R19	HAZOP	Add an independent low-low level on V-104 closing a shutdown valve in the product line; check the TK-106 / TK-107 vents for gas blow-by.	SIS	PK-10
R21	HAZOP	Add a PI-101 high alarm between the PIC-105 alarm and PSHH-101, because PIC-105 measures at V-104, not at R-101.	SIS	PK-11
R22	HAZOP	Add a high alarm on TI-109 (V-104 temperature).	SIS	PK-11
R23	HAZOP	Add a low-flow alarm on the purge FI-107 and a purity check in the shift routine.	SIS	PK-11
R24	HAZOP	Add low-low hydrogen flow (FI-103) to the SIS to close XV-101; today only a stopped K-101 motor is seen.	SIS	PK-10
R25	HAZOP	Install hydrogen gas detection at K-101, the reactors and V-104, and alarm the loop hydrogen balance (make-up against 3 × benzene fed).	HW	PK-17
R26	HAZOP	Add a second loop oxygen analyser (2oo2) for the XV-105 permissive and record the O2 readings in the purge procedure.	SIS	PK-10
R27	HAZOP	Fit a non-return valve on the K-101 discharge and confirm the kickback / anti-surge protection and its high-temperature alarm.	HW	PK-07
R28	HAZOP	Protect the nitrogen connection: non-return valve, double block and bleed or a removable spool; HV-120 opened only below the header pressure.	HW	PK-14
R29	HAZOP	Depressuring study for EDV-101 (rate, flare load, minimum metal temperature); fail-open actuator and proof test.	DOC	PK-10
R30	HAZOP	Provide a flare knock-out drum and a continuous flare purge or seal to keep air out.	HW	PK-14
R32	HAZOP	Light-off and shutdown steps: limit the feed ramp after XV-101 opens, stop benzene before hydrogen, and add a low-temperature-with-feed interlock (TI-100 with XV-101 open) to stop benzene accumulation.	DOC	PK-10
R33	HAZOP	Add a TIC-101 against TI-100 deviation alarm (two elements in the same reactor).	SIS	PK-11
R35	HAZOP	Fit minimum-flow bypasses on P-100A/B and P-102A/B.	HW	PK-06
R37	HAZOP	Add an R-103 bed-rise alarm (TI-107E minus TI-106) as early warning of benzene slip.	SIS	PK-11
R38	HAZOP	Positive segregation of MP steam and CW on the R-102 jacket (interlocked change-over) and jacket relief.	HW	PK-14
SL-R1	Safety layers	R-101 pressure pre-alarm on PT-101 (PIAH-101, H 32.5 bar g (assumed))	SIS	PK-11

ID	Source	Recommendation	Disposition	Package
SL-R2	Safety layers	Deviation alarm TIC-101 element vs TI-100 element (TDA-101, |ΔT| > 5 °C (assumed))	SIS	PK-11
SL-R3	Safety layers	D-101 level low-low → unit trip (LSLL-109, 10 % (assumed))	SIS	PK-10
SL-R4	Safety layers	Agitator shaft-speed switch, voted with the M-101 run signal (SSL-101, low speed (assumed 50 % of design))	SIS	PK-10
SL-R5	Safety layers	Separate SIS elements in R-103 (two per cell, 2oo3 per hot zone) (TSHH-107S, 220 °C)	SIS	PK-10
SL-R6	Safety layers	V-104 level low-low closes a new SIS valve XV-104 on the product line (LSLL-104, 5 % (assumed))	SIS	PK-10
SL-R7	Safety layers	Stop P-100A/B on unit trip (I-P100, on every unit trip)	SIS	PK-10
SL-R8	Safety layers	Safety valve on the R-101 overhead (no isolation between it and R-101) (PSV-101, 36 bar g (assumed design pressure))	HW	PK-15
SL-R9	Safety layers	Safety valve on the R-102 off-gas line (PSV-102, 36 bar g (assumed))	HW	PK-15
SL-R10	Safety layers	Safety valve on V-104 (PSV-104, 36 bar g (assumed))	HW	PK-15
SL-R11	Safety layers	Safety valve on K-101 discharge (upstream of FV-103) (PSV-112, 42 bar g (assumed))	HW	PK-15
SL-R12	Safety layers	Safety valve on steam drum D-101 (PSV-109, 8 bar g (assumed))	HW	PK-15
SL-R13	Safety layers	Thermal relief on the E-103 cooling-water side (TSV-113, 10 bar g (assumed))	HW	PK-15
SL-R14	Safety layers	Thermal relief on the E-106 cooling-water side (TSV-106, 10 bar g (assumed))	HW	PK-15
SL-R15	Safety layers	Thermal relief on the R-102 jacket (TSV-102J, 10 bar g (assumed))	HW	PK-15
SL-R16	Safety layers	Thermal relief on each catalyst filter F-102A/B (TSV-102F, 36 bar g (assumed))	HW	PK-15
SL-R17	Safety layers	Emergency isolation valve on benzene at battery limit (EIV-100, fail closed, fire-safe)	HW	PK-15
SL-R18	Safety layers	EIV on the R-101 bottom outlet (pump-around suction) (EIV-101, fail closed, fire-safe)	HW	PK-15
SL-R19	Safety layers	EIV on the R-102 bottom outlet (EIV-102, fail closed, fire-safe)	HW	PK-15
SL-R20	Safety layers	EIV on the V-104 product line (EIV-104, fail closed, fire-safe)	HW	PK-15
SL-R21	Safety layers	Check valve at P-100A/B discharge (NRV-100)	HW	PK-15
SL-R22	Safety layers	Check valve at K-101 discharge (NRV-112)	HW	PK-07
SL-R23	Safety layers	Flare knock-out drum (V-190)	HW	PK-14
SL-R24	Safety layers	Kerbed area round the benzene pumps to the oily-water sewer (BUND-1)	HW	PK-14
SL-R25	Safety layers	Kerbed slab under the reactor structure to a closed drain (BUND-2)	HW	PK-14
SL-R26	Safety layers	Raney-Ni handled under liquid and nitrogen (catalyst pot) (CAT-1, N2 blanket)	DOC	PK-13
SL-R27	Safety layers	Spent catalyst from F-102 kept wet in closed drums (CAT-2)	DOC	PK-13
SL-R28	Safety layers	Fireproofing of the R-103 skirt and V-104 saddles (FP-1, 2 h (assumed))	HW	PK-14
SL-R29	Safety layers	Fire-water ring main (FW-1, 10 bar g (assumed))	HW	PK-17
SL-R30	Safety layers	Hydrant west of the reactors (HYD-1)	HW	PK-17
SL-R31	Safety layers	Hydrant east of V-104 (HYD-2)	HW	PK-17
SL-R32	Safety layers	Fire-water monitor on the reactor structure (MON-1)	HW	PK-17
SL-R33	Safety layers	Fire-water monitor at V-104 / K-101 (MON-2)	HW	PK-17
SL-R34	Safety layers	Deluge on K-101 (DEL-101, opens on confirmed fire (assumed))	HW	PK-17
SL-R35	Safety layers	Water spray on V-104 (DEL-104, opens on confirmed fire (assumed))	HW	PK-17
SL-R36	Safety layers	Hydrogen detector at K-101 (GD-1, 10 % / 25 % LFL (assumed))	HW	PK-17
SL-R37	Safety layers	Hydrogen detector at the R-101 top flanges (GD-2, 10 % / 25 % LFL (assumed))	HW	PK-17

ID	Source	Recommendation	Disposition	Package
SL-R38	Safety layers	Hydrocarbon (benzene) detector at P-100A/B (GD-3, 10 % / 25 % LFL (assumed))	HW	PK-17
SL-R39	Safety layers	UV/IR flame detector covering K-101 and V-104 (FD-1)	HW	PK-17
SL-R40	Safety layers	Manual call point (MCP-1)	HW	PK-17
SL-R41	Safety layers	Site siren / PA (SIREN)	HW	PK-17
SL-R42	Safety layers	Windsock (WIND)	HW	PK-17
SL-R43	Safety layers	Muster point (MUSTER)	HW	PK-17
SL-R44	Safety layers	Safety shower and eye wash at the benzene pumps (SHOWER)	HW	PK-17

7 · Engineering queries

Inconsistencies found while writing this document and the commissioning plan; they are collected in the document register (Project & deliverables tab).

EQ	Query	Documents	Proposed resolution
EQ-ec-1	R-101 and R-102 are both slurry reactors, but the model has an agitator motor on R-101 only (MOTORS: M-101) and the P&ID shows no mixer on R-102. How R-102 keeps the Raney-Ni suspended is not defined.	PK-01; equipment list CHX-V1-ME-EQL-001; load list CHX-V1-EE-LL-001	Process to state the R-102 mixing (agitator or gas-induced). If agitated, add M-102 to PK-01, the load list and the motor solo runs.
EQ-ec-2	In HEAT mode (HS-101) MP steam at 10 bar g reaches the E-101 shell, which is the D-101 steam side protected by PSV-109 at an assumed 8.0 bar g (Safety layers SL-R12). The shell, the drum and TP-05 have no agreed design pressure.	PK-03, PK-04, TP-05; Safety layers SL-R12; relief load basis CHX-V1-PE-RLB-001	Either design the E-101 shell and D-101 for the MP steam pressure (PSV-109 set to suit) or add pressure reduction and relief on the MP steam to E-101. Then fix TP-05.
EQ-ec-3	The Raney-Ni catalyst pot on R-101 (HV-117) has no equipment number on the P&ID and no datasheet, yet SOP-CYC-07 pressures it with nitrogen above the R-101 pressure.	P&ID CHX-V1-PID-001; PK-05; SOP-CYC-07	Give the pot an equipment number and a datasheet; design pressure at least the loop design pressure; add it to test pack TP-01.
EQ-ec-4	The SIS scope is not frozen. Open recommendations add sensors, logic and valves: HAZOP R01, R03–R07, R14, R16, R19, R24 and R26; Safety layers SL-R3 … SL-R7; the second make-up valve of the LOPA and SRS.	PK-10; C&E CHX-V1-IN-CE-001; SRS CHX-V1-IN-SRS-001	Close these in the HAZOP close-out before the SIS PO (week 18 in part 4). Order with spare I/O (about 20 %, typical). Re-issue the C&E, then the SIS FAT procedure.
EQ-ec-5	Raney-Ni usually comes water-wet. SOP-CYC-07 rinses with cyclohexane but has no step to replace the water. Water charged into R-101 at 200 °C flashes, then ends up in V-104 and the product.	PK-13; SOP-CYC-07; checklist CL-03	Specify the carrier liquid in PK-13: either supplied under cyclohexane, or a solvent exchange in the pot as a written step in SOP-CYC-07.
EQ-ec-6	The relief valves (SL-R8 … SL-R16) are recommended additions with assumed set pressures and no orifice sizes. PK-15 cannot be bought, and MC-04 and PSSR item PS-03 cannot be closed, until the relief load basis is issued.	PK-15; MC-04; PS-03; Relief & vent tab CHX-V1-PE-RLB-001	Issue the relief load basis before week 24 (the PK-15 PO week); set pressures follow the design pressures in EQ-ec-2.
EQ-ec-7	The design pressure of the benzene feed line (TP-04) is assumed equal to the loop (36 bar g). The P-100A/B shut-off head with a closed outlet may be higher.	PK-06; TP-04; line list CHX-V1-PE-LL-001	Take the P-100 shut-off pressure from the pump datasheet. Set the line design pressure, and the TP-04 test pressure, from it.